

Chapter 3

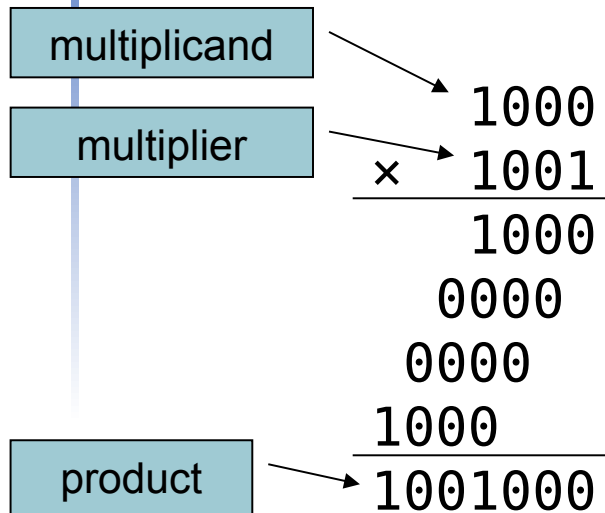
Arithmetic for Computers

Arithmetic for Computers

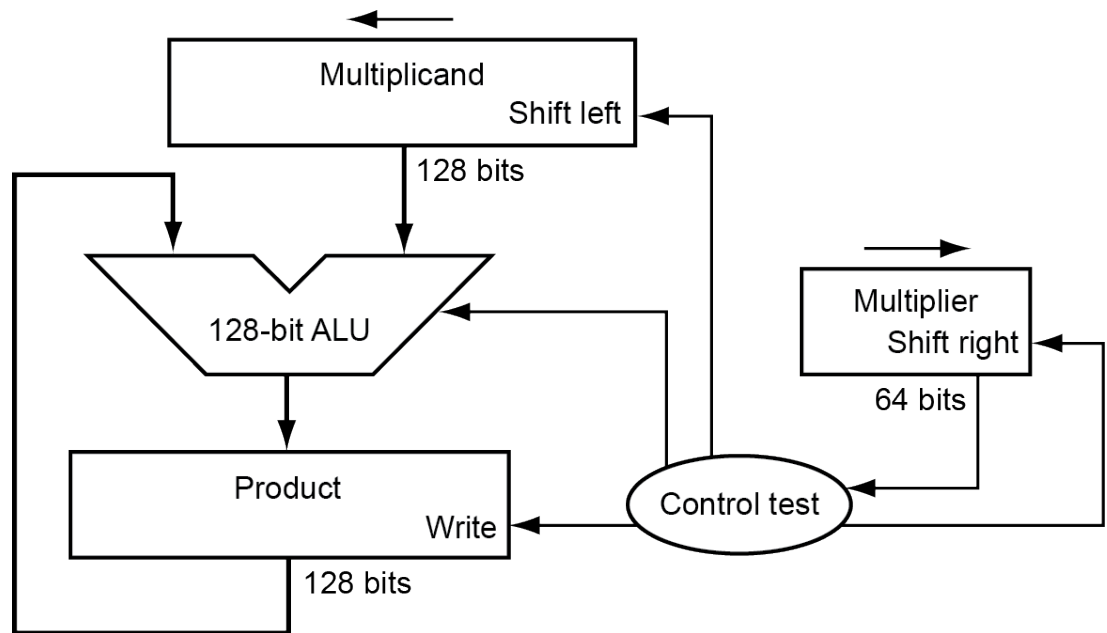
- Operations on integers
 - Addition and subtraction
 - Multiplication and division
 - Dealing with overflow
- Floating-point real numbers
 - Representation and operations

Multiplication

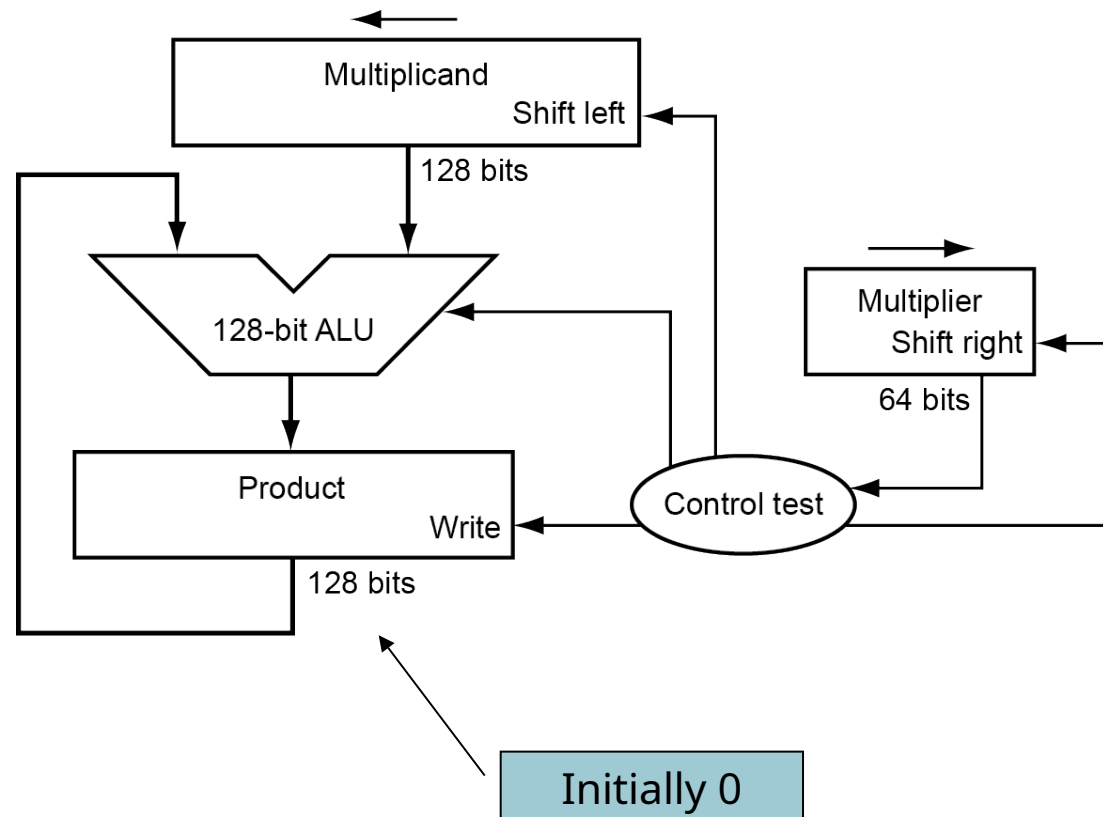
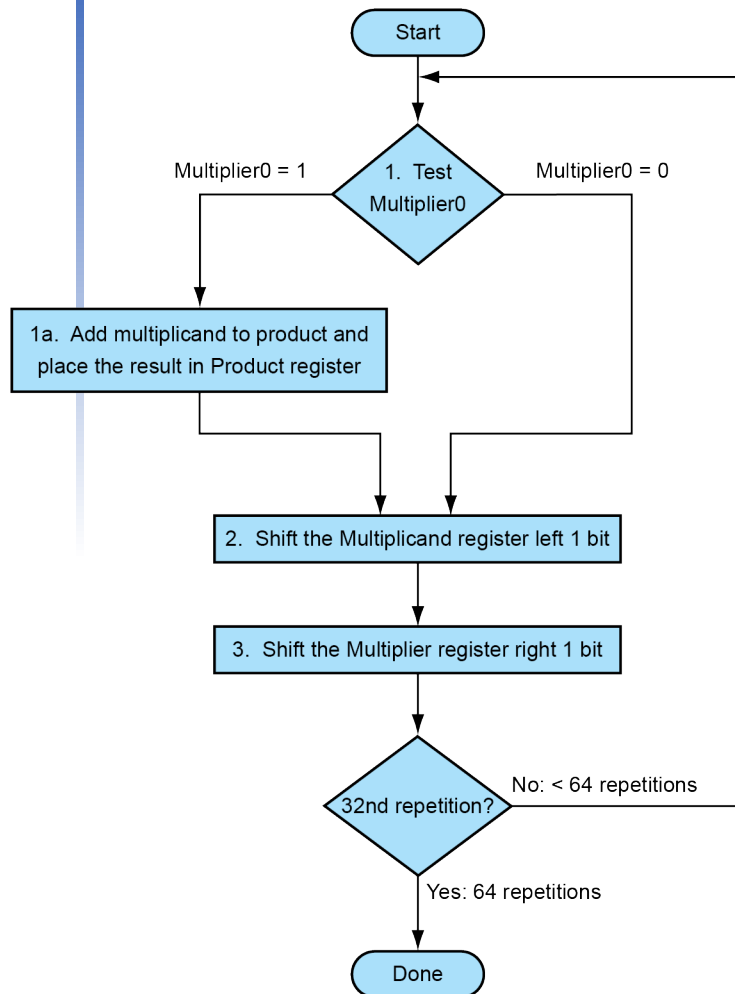
- Start with long-multiplication approach



Length of product is the sum of operand lengths



Multiplication Hardware



Floating Point

- Representation for non-integral numbers
 - Including very small and very large numbers
- Like scientific notation
 - -2.34×10^{56} ← normalized
 - $+0.002 \times 10^{-4}$ ← not normalized
 - $+987.02 \times 10^9$ ← not normalized
- In binary
 - $\pm 1.xxxxxxx_2 \times 2^{yyyy}$
- Types `float` and `double` in C

Floating Point Standard

- Defined by IEEE Std 754-1985
- Developed in response to divergence of representations
 - Portability issues for scientific code
- Now almost universally adopted
- Two representations
 - Single precision (32-bit)
 - Double precision (64-bit)

IEEE Floating-Point Format

single: 8 bits

double: 11 bits

single: 23 bits

double: 52 bits

S	Exponent	Fraction
---	----------	----------

$$x = (-1)^S \times (1 + \text{Fraction}) \times 2^{(\text{Exponent} - \text{Bias})}$$

- S: sign bit (0 $\Rightarrow$ non-negative, 1 $\Rightarrow$ negative)
- Normalize significand: $1.0 \leq |\text{significand}| < 2.0$
 - Always has a leading pre-binary-point 1 bit, so no need to represent it explicitly (hidden bit)
 - Significand is Fraction with the “1.” restored
- Exponent: excess representation: actual exponent + Bias
 - Ensures exponent is unsigned
 - Single: Bias = 127; Double: Bias = 1203

Single-Precision Range

- Exponents 00000000 and 11111111 reserved
- Smallest value
 - Exponent: 00000001
 $\Rightarrow$ actual exponent = $1 - 127 = -126$
 - Fraction: 000...00 $\Rightarrow$ significand = 1.0
 - $\pm 1.0 \times 2^{-126} \approx \pm 1.2 \times 10^{-38}$
- Largest value
 - exponent: 11111110
 $\Rightarrow$ actual exponent = $254 - 127 = +127$
 - Fraction: 111...11 $\Rightarrow$ significand ≈ 2.0
 - $\pm 2.0 \times 2^{+127} \approx \pm 3.4 \times 10^{+38}$

Double-Precision Range

- Exponents 0000...00 and 1111...11 reserved
- Smallest value
 - Exponent: 000000000001
 $\Rightarrow$ actual exponent = $1 - 1023 = -1022$
 - Fraction: 000...00 $\Rightarrow$ significand = 1.0
 - $\pm 1.0 \times 2^{-1022} \approx \pm 2.2 \times 10^{-308}$
- Largest value
 - Exponent: 111111111110
 $\Rightarrow$ actual exponent = $2046 - 1023 = +1023$
 - Fraction: 111...11 $\Rightarrow$ significand ≈ 2.0
 - $\pm 2.0 \times 2^{+1023} \approx \pm 1.8 \times 10^{+308}$

Floating-Point Precision

- Relative precision
 - all fraction bits are significant
 - Single: approx 2^{-23}
 - Equivalent to $23 \times \log_{10} 2 \approx 23 \times 0.3 \approx 6$ decimal digits of precision
 - Double: approx 2^{-52}
 - Equivalent to $52 \times \log_{10} 2 \approx 52 \times 0.3 \approx 16$ decimal digits of precision

Floating-Point Example

- Represent -0.75
 - $-0.75 = (-1)^1 \times 1.1_2 \times 2^{-1}$
 - $S = 1$
 - Fraction = $1000\dots00_2$
 - Exponent = $-1 + \text{Bias}$
 - Single: $-1 + 127 = 126 = 01111110_2$
 - Double: $-1 + 1023 = 1022 = 011111111110_2$
- Single: $10111111101000\dots00$
- Double: $101111111111101000\dots00$

Floating-Point Example

- What number is represented by the single-precision float

11000000101000...00

- $S = 1$

- Fraction = $01000...00_2$

- Exponent = $10000001_2 = 129$

- $$\begin{aligned}x &= (-1)^1 \times (1 + 01_2) \times 2^{(129 - 127)} \\&= (-1) \times 1.25 \times 2^2 \\&= -5.0\end{aligned}$$

Denormal Numbers

- Exponent = 000...0 $\Rightarrow$ hidden bit is 0

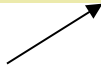
$$x = (-1)^S \times (0 + \text{Fraction}) \times 2^{-\text{Bias}}$$

- Smaller than normal numbers
 - allow for gradual underflow, with diminishing precision

- Denormal with fraction = 000...0

$$x = (-1)^S \times (0 + 0) \times 2^{-\text{Bias}} = \pm 0.0$$

Two representations
of 0.0!



Infinites and NaNs

- Exponent = 111...1, Fraction = 000...0
 - $\pm$ Infinity
 - Can be used in subsequent calculations, avoiding need for overflow check
- Exponent = 111...1, Fraction $\neq$ 000...0
 - Not-a-Number (NaN)
 - Indicates illegal or undefined result
 - e.g., 0.0 / 0.0
 - Can be used in subsequent calculations

Concluding Remarks

- Bits have no inherent meaning
 - Interpretation depends on the instructions applied
- Computer representations of numbers
 - Finite range and precision
 - Need to account for this in programs

Concluding Remarks

- ISAs support arithmetic
 - Signed and unsigned integers
 - Floating-point approximation to reals
- Bounded range and precision
 - Operations can overflow and underflow